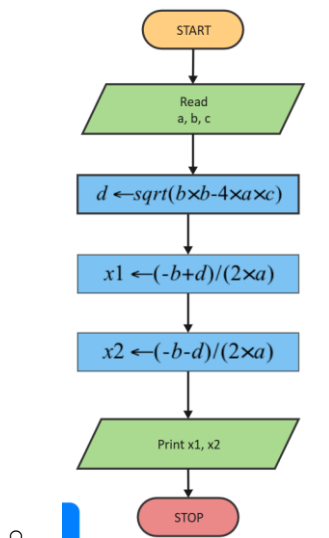


2.1.1. Roots of a Quadratic Equation

Algorithm: Input: Three integers a, b, c (coefficients of $ax^2 + bx + c = 0$)

Output: Roots of the equation formatted according to their type

1. Start
2. Read the coefficients a, b, c
3. Calculate the discriminant:
 $D = b^2 - 4ac$
4. If $D > 0$ (discriminant positive)
 - Roots are real and different
 - Calculate $root1 = (-b + \sqrt{D}) / (2a)$
 - Calculate $root2 = (-b - \sqrt{D}) / (2a)$
 - Print:
 - Flowchart:



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2.1.1. Roots of a Quadratic Equation

The discriminant $D = b^2 - 4ac$ determines the nature of the roots:

- If $D > 0$: Roots are real and different
- If $D = 0$: Roots are real and the same
- If $D < 0$: Roots are imaginary

Input Format:

- Three space-separated integers representing the coefficients a, b, and c, respectively.

Output Format:

- If roots are real and different, print:
root1 = <root1>
root2 = <root2>
- If roots are the same, print:
root1 = root2 = <root1>
- If roots are imaginary, print:
root1 = <RealPart>+<ImaginaryPart>i
root2 = <RealPart>-<ImaginaryPart>i
- All values should be formatted to two decimal places.

Sample Test Cases

```
1 # write your code here
2 a, b, c = map(float, input().split())
3 D = (b*b) - (4*a*c)
4 sqrtD = D ** 0.5
5 root1 = (-b+sqrtD)/(2*a)
6 root2 = (-b-sqrtD)/(2*a)
7 if D > 0:
8     print(f"root1 = {root1:.2f}")
9     print(f"root2 = {root2:.2f}")
10 elif D == 0:
11     print(f"root1 = root2 = {root1:.2f}")
12 else:
13     pass
```

Average time: 0.012 s, Maximum time: 0.030 s, 3 out of 3 shown test case(s) passed, 3 out of 3 hidden test case(s) passed

Test case 1: Expected output: 3.00, 2.00, Actual output: 3.00, 2.00

Test case 2: (1 ms)